Course name: Oceanography and Limnology
Course code: 2103
Lecture no.- 01
Lecture on: Major Oceans, Bay of Bengal & Bangladesh Maritime areas

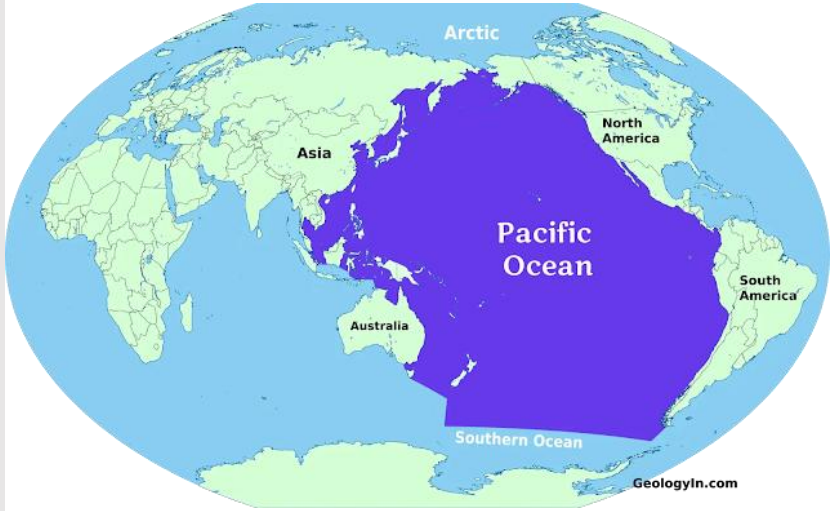
5 major oceans

- Oceans play critical roles in **global climate regulation, trade, and biodiversity**.



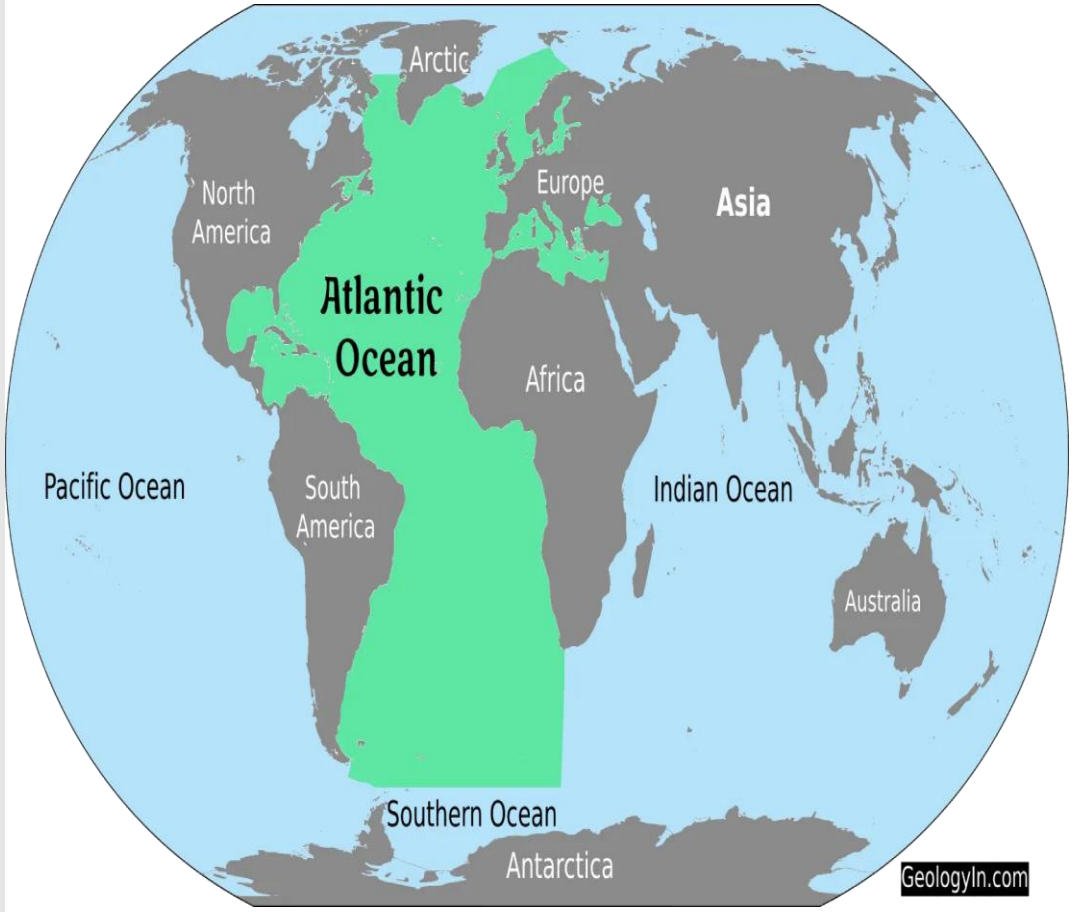
1. Pacific Ocean

- **Size**: The **largest ocean**, covering about **63 million square miles (165 million square kilometers**), making up **more than one-third of Earth's surface**.
- **Location**: Situated between **Asia and Australia on the west**, and the **Americas on the east**.
- **Depth**: The average depth is around 12,080 feet (**3,682 meters**), with the Mariana Trench being the **deepest point at about 36,070 feet** (10,994 meters).
- **Key Features**: Known for its **vast expanse**, the Pacific Ocean has **numerous islands** and is prone to strong seismic activity like **earthquakes and tsunamis**. It's also home to the **Great Barrier Reef** and various other marine ecosystems.



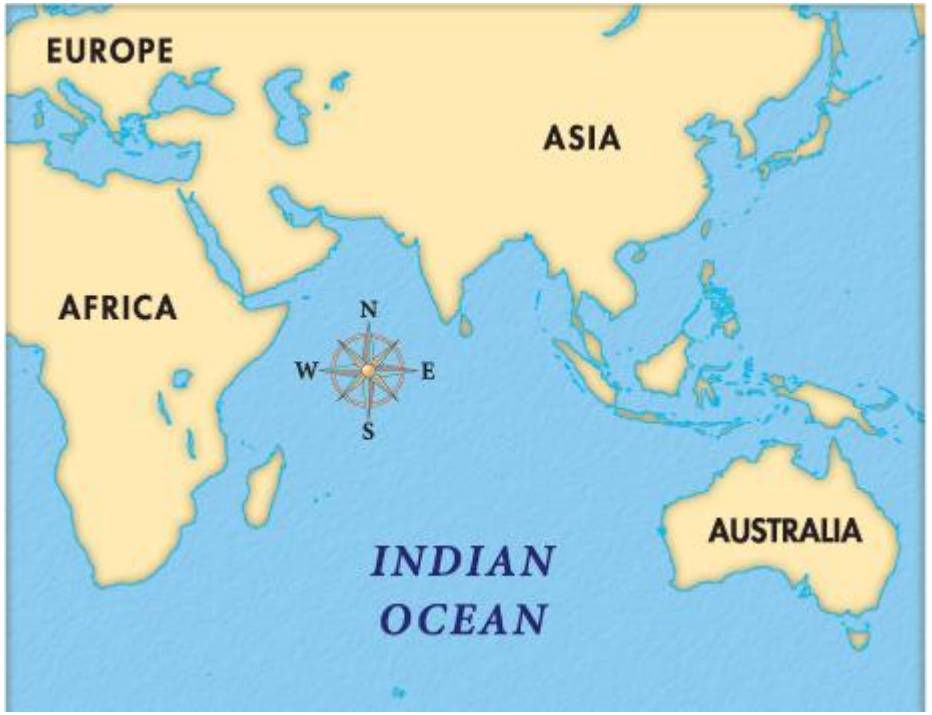
2. Atlantic Ocean

- **Size**: The **second-largest ocean**, covering approximately **41 million square miles** (106 million square kilometers).
- **Location**: Lies between the **Americas on the west** and **Europe and Africa on the east**.
- **Depth**: The average depth is 10,930 feet (**3,332 meters**), with the **Puerto Rico Trench** being the **deepest part** at about 28,232 feet (**8,605 meters**).
- **Key Features**: The Atlantic is crucial for **international trade routes** and features the **mid-Atlantic ridge**, a **major underwater mountain** range. The **Bermuda Triangle**, which has been associated with **mysterious disappearances**, is located in this ocean.



3. Indian Ocean

- **Size**: The **third-largest ocean**, covering about 27 million square miles (**70 million square kilometers**).
- **Location**: Bounded by **Asia to the north, Africa to the west, Australia to the east, and the Southern Ocean to the south.**
- **Depth**: The average depth is about 12,080 feet (**3,682 meters**), with the **Sunda Trench** being the deepest point at around 24,442 feet (**7,455 meters**).
- **Key Features**: The Indian Ocean **plays a significant role in global climate systems** (e.g., **the monsoon**). It is important for **global shipping routes** and is surrounded by several **key economic regions** like the Arabian Peninsula, India, and East Africa.



4. Southern Ocean (Antarctic Ocean)

- **Size**: Covers about **7.8 million square miles** (20 million square kilometers), making it the **smallest** of the five oceans.
- **Location**: Surrounds **Antarctica**, extending northward to the **60°S latitude**.
- **Depth**: The average depth is 13,100 feet (**4,000** meters), with the **South Sandwich Trench** being the deepest point at about 23,737 feet (**7,235 meters**).
- **Key Features**: Known for its **extreme cold**, the Southern Ocean is **vital for regulating Earth's climate** through ocean currents. It is **rich in marine life**, including **krill**, which forms the base of the food chain in the region.



5. Arctic Ocean

- **Size**: The **smallest ocean**, covering about 5.4 million square miles (**14 million square kilometers**).
- **Location**: Found mostly within the Arctic Circle, it borders the **northern parts of North America**, Europe, and Asia.
- **Depth**: The average depth is about 3,953 feet (**1,205 meters**), with the Fram Basin being the **deepest point** at around 17,880 feet (**5,450 meters**).
- **Key Features**: The Arctic Ocean is characterized by **sea ice that fluctuates with the seasons**. It's one of the **least explored oceans**, but with climate change, there has been **increasing interest in its natural resources and shipping routes.**



Summary Table of the Five Major Oceans

Feature	Pacific Ocean	Atlantic Ocean	Indian Ocean	Southern Ocean	Arctic Ocean
Approx. area (million km²)	165.25	106.46	70.56	21.96	15.56
Approx. average depth (m)	4,000	3,646	3,741	3,270	1,205
Maximum depth (m)	10,984	8,376	7,258	7,432	5,550
Location	Between Asia, Australia, and the Americas	Between the Americas, Europe, and Africa	Between Africa, Asia, Australia, and Antarctica	Surrounds Antarctica	Around the North Pole
Largest marginal sea / notable feature	Philippine Sea	Caribbean Sea	Arabian Sea	Weddell Sea	Barents Sea
Major currents	Kuroshio, California, Humboldt	Gulf Stream, Labrador, Canary	Agulhas, Somali, West Australian	Antarctic Circumpolar Current	Beaufort Gyre, Transpolar Drift
Climate characteristics	Tropical to polar	Tropical to polar	Predominantly tropical and subtropical	Polar and subpolar	Polar
Key importance	Fisheries, trade, biodiversity	Global trade, fisheries, climate regulation	Monsoon systems, fisheries, shipping	Global ocean circulation and carbon uptake	Sea-ice regulation and climate feedback
Approx. share of global ocean area	46%	29%	20%	6%	4%

Bay of Bengal

- ✓ It is located between latitudes **5°N and 22°N** and longitudes **80°E and 100°E**.
- ✓ It is a **northern extension** of the Indian Ocean, positioned between **India and Sri Lanka** in the west, Bangladesh to the **north**, and Myanmar (Burma) and the northern part of the Malay Peninsula to the **east**.
- ✓ It is bounded in the **west** by the east coasts of **Sri Lanka and India,** on the **north** by the deltaic region of the **Ganges-Brahmaputra-Meghna river system**, and on the **east** by the Myanmar peninsula extended up to the **Andaman-Nicobar ridges.**
- ✓ The southern boundary of the Bay is approximately along the line drawn from **Dondra Head** in the south of Sri Lanka to the **north tip of Sumatra**.
- ✓ The Bay occupies an area of about **2.2 million sq km.**



Bay of Bengal

- ✓ It is approximately **1,000 miles (1,600 km) wide**, with an average depth near 8,500 feet (**2,600 meters**). The maximum depth is recorded at 15,400 feet (**4,694 meters**).
- ✓ Bangladesh is situated at the **head** of the Bay of Bengal.
- ✓ **Fisheries production** in the Bay of Bengal is **six million tonnes per year**, more than **7%** of the world's catch.
- ✓ The Bay of Bengal is an area of high biodiversity, with a large number of **endangered and vulnerable species**.
- ✓ Bay of Bengal is **the 64th largest** marine Ecosystem.



Summary

Feature	Description
Location	Northeastern part of the Indian Ocean
Approximate area	2.17 million km²
Approximate length	2,090 km (north–south)
Approximate width	1,610 km (east–west)
Average depth	Approximately 2,600 m
Maximum depth	Approximately 4,694 m
Northern boundary	Bangladesh and the eastern coast of India
Western boundary	Eastern coast of India
Eastern boundary	Myanmar and the Andaman–Nicobar Islands
Southern boundary	Open Indian Ocean
Major river inputs	Ganges–Brahmaputra–Meghna, Irrawaddy, Mahanadi, Godavari, Krishna, and other rivers
Major countries bordering the bay	Bangladesh, India, Myanmar, and Sri Lanka (near its western approach)
Major islands	Andaman and Nicobar Islands, St. Martin's Island
Major currents	East India Coastal Current, West India Coastal Current, and seasonally reversing circulation
Sea surface temperature (SST)	Generally around 26–30°C, with seasonal and regional variations
Salinity	Relatively low in the northern bay due to heavy rainfall and river discharge; generally higher toward the south
Climate influence	South Asian monsoon, tropical cyclones, and regional rainfall variability
Major hazards	Tropical cyclones, storm surges, coastal erosion, flooding, and sea-level rise
Ecological importance	Mangroves, coral communities, estuaries, fisheries, and marine biodiversity
Economic importance	Fisheries, maritime transport, offshore energy, ports, and coastal livelihoods
Major environmental concerns	Plastic pollution, heavy-metal contamination, eutrophication, habitat degradation, and ocean warming

United Nations Conference on the Law of the Sea (UNCLOS)

The United Nations Conference on the Law of the Sea (UNCLOS) was conducted in three major rounds to establish international rules governing ocean use, maritime boundaries, and marine resources.

Feature	UNCLOS I	UNCLOS II	UNCLOS III
Full name	First United Nations Conference on the Law of the Sea	Second United Nations Conference on the Law of the Sea	Third United Nations Conference on the Law of the Sea
Period	1958	1960	1973–1982
Location	Geneva, Switzerland	Geneva, Switzerland	Multiple sessions in New York, Caracas, Geneva, and other locations
Main objective	Establish international rules for the law of the sea	Resolve unresolved maritime issues from UNCLOS I	Develop a comprehensive legal framework for the oceans
Major outcome	Four Geneva Conventions on the Law of the Sea	No major treaty adopted	United Nations Convention on the Law of the Sea (UNCLOS)
Territorial sea	No universally agreed breadth	Failed to agree on a uniform breadth	Established a maximum breadth of 12 nautical miles
Contiguous zone	Recognized, generally up to 12 nautical miles from the baseline	No major agreement on the breadth of the territorial sea	Up to 24 nautical miles from the baseline
Continental shelf	Convention on the Continental Shelf adopted	No major new agreement	Comprehensive rules for continental shelf rights, including possible extension beyond 200 nautical miles
High seas	Convention on the High Seas adopted	No major agreement	Comprehensive provisions on navigation, resources, and other uses
Fisheries	Convention on Fishing and Conservation of the Living Resources of the High Seas adopted	Failed to resolve key disputes	Framework for fisheries management and conservation
Dispute settlement	Convention on the Settlement of Disputes adopted	No major new framework	Detailed dispute-settlement procedures
Overall significance	Established foundational maritime law	Demonstrated the need for broader agreement	Created the modern, comprehensive law of the sea

United Nations Conference on the Law of the Sea (UNCLOS)

❑ **Four Conventions adopted under UNCLOS I (1958)**

1.Convention on the Territorial Sea and the Contiguous Zone.

2.Convention on the High Seas.

3.Convention on Fishing and Conservation of the Living Resources of the High Seas.

4.Convention on the Continental Shelf.

A fifth related instrument, the Optional Protocol of Signature concerning the Compulsory Settlement of Disputes, was also adopted.

❑ **Key facts about UNCLOS III**

•Adopted: 10 December 1982, in Montego Bay, Jamaica.

•Entered into force: 16 November 1994.

•Territorial sea: Up to 12 nautical miles.

•Exclusive Economic Zone (EEZ): Up to 200 nautical miles.

•**Continental shelf:** At least 200 nautical miles where applicable, with possible extension under the Convention's criteria.

•**International Seabed Area:** The seabed beyond national jurisdiction is governed by a specific international legal regime.

❑ **Importance for Bangladesh:** UNCLOS provides the legal framework for maritime delimitation, fisheries management, offshore energy exploration, marine environmental protection, and the country's maritime rights in the Bay of Bengal.

Recent Achievement of Bangladesh on the Right on Sea

❑ Introduction:

- Bangladesh is a coastal state being surrounded by another two coastal states **India and Myanmar**. Since the emergence as an **independent country in 1971**, the country has **maritime boundary dispute** with its two neighbor countries. Recently Bangladesh govemment had won a huge area from the **two neighbor countries India and Myanmar**, which is known as **"Somudro Bijoy".**

❑ Background:

- Myanmar maritime boundary dispute with Bangladesh and India in particular **began in 1974**. At that time **Territorial Waters and Maritime Zones Act"** was passed in parliament. By law, the government announced that the country's maritime boundary. The border with India and Myanmar and they had **protested**. At the same time, they **refused to accept** the border. Bangladesh would be the **proper way**; they did not define its borders. Since then, the large part of the Bay of Bengal has survived as a disputed territory. Bangladesh **last 38 years** to resolve the dispute with the two countries have discussed on numerous occasions, but there **has not been any progress**. Consequently, the country has to go to **the international court**.
- Settlement of the maritime dispute with **Myanmar took 2009** decisions to the International Court of Bangladesh in October. Accordingly, as of **December 14, ITLOS 16th case** was included in the case. Myanmar has **agreed to accept the decision** of the judicial authorities, however, **took part in ITLOS**.
- In 1 July 2010 Bangladesh and 1 December Myanmar **presented documentation** for the first time. The argument **refuted** by Myanmar 2011 on March 15, their argument highlights Bangladesh and Myanmar on **July 1 last year**, the arguments to refute the arguments presented.
- ITLOS gave **the verdict**. The message has been **sent to the Bangladesh Embassy in Germany**, recognized **211 votes in the 151-page verdict** read ITLOS president **Jose Luis Jesus**. Roy between Bangladesh and Myanmar in the Bay of **oil-gas long unresolved border** is finally determined. At the same time, the two close neighbors to determine the boundaries of the Bay of Bengal for **more than three decades**, has long disputed settlement. Some two-year dispute over the maritime arbitration court verdict was given.

❑ Significant gains for Bangladesh in the Maritime Boundary Dispute with Myanmar:

- The Bay of Bengal had **demanded 107,000 square kilometres**, where the court has determined **111,000 square kilometres**. Bangladesh is now in the **oil and gas exploration in the Bay of Bengal** will begin the process of working. Now, **354 nautical miles of the continental shelf** in the Bay of Bengal, the ruling of the claim has been established.

Significant gains for Bangladesh in the Maritime Boundary Dispute with India

- ✓ Bangladesh has been gained **19,467 square kilometres out of 25,602 sq km** disputed area in the Bay of Bengal.
- ✓ The **19,467 square kilometres** of Bangladesh's territorial water had long been caught up in dispute with India.
- ✓ The government went in for arbitration over the delimitation of sea boundary under the United Nations Convention on Law of Sea **(UNCLOS) on October 8, 2009.** The hearings ended in December last year when both sides argued their case before the **PCA in the Netherlands capital**.
- ✓ The argument focused on issues including the location of the land boundary terminus, the **delimitation of the territorial sea, the exclusive economic zone, and the continental shelf** within and beyond 200 nautical miles.
- ✓ Meanwhile, in **2012, a 21-member** Intenational Tribunal for the Law of the Sea (**ITLOS**) in Hamburg issued judgment **rejecting equidistance' boundaries** (as proposed by Myanmar in that case) and awarded Bangladesh a sizable share of the disputed waters in the eastern part of the Bay of Bengal, including the **continental shelf beyond 200nm**.



Fig: Bangladesh has gained **19,467** square kilometers of maritime areas at the Bay of Bengal out of the disputed **25,602** square kilometers with **neighboring India.**

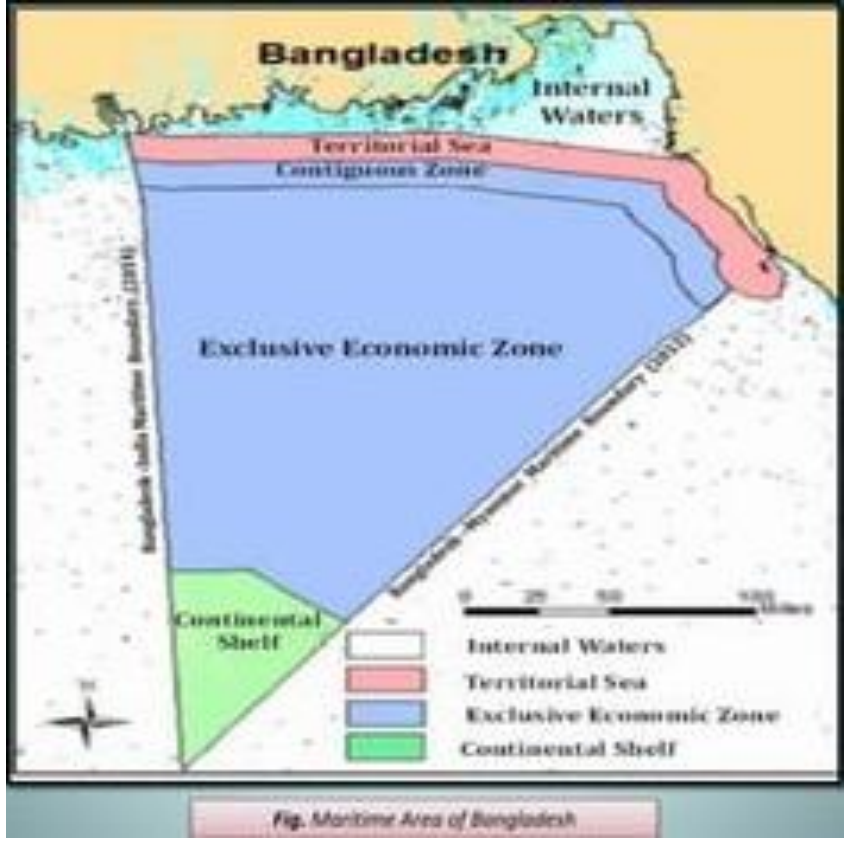
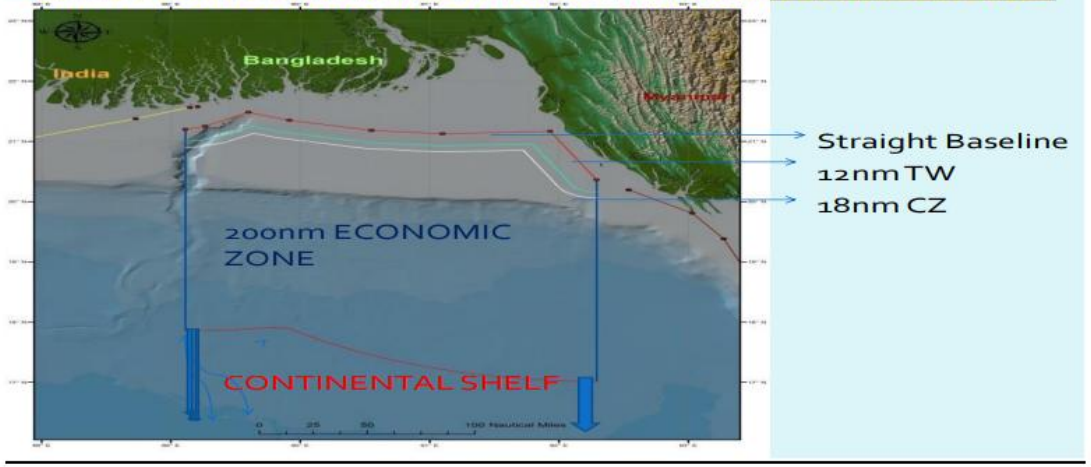


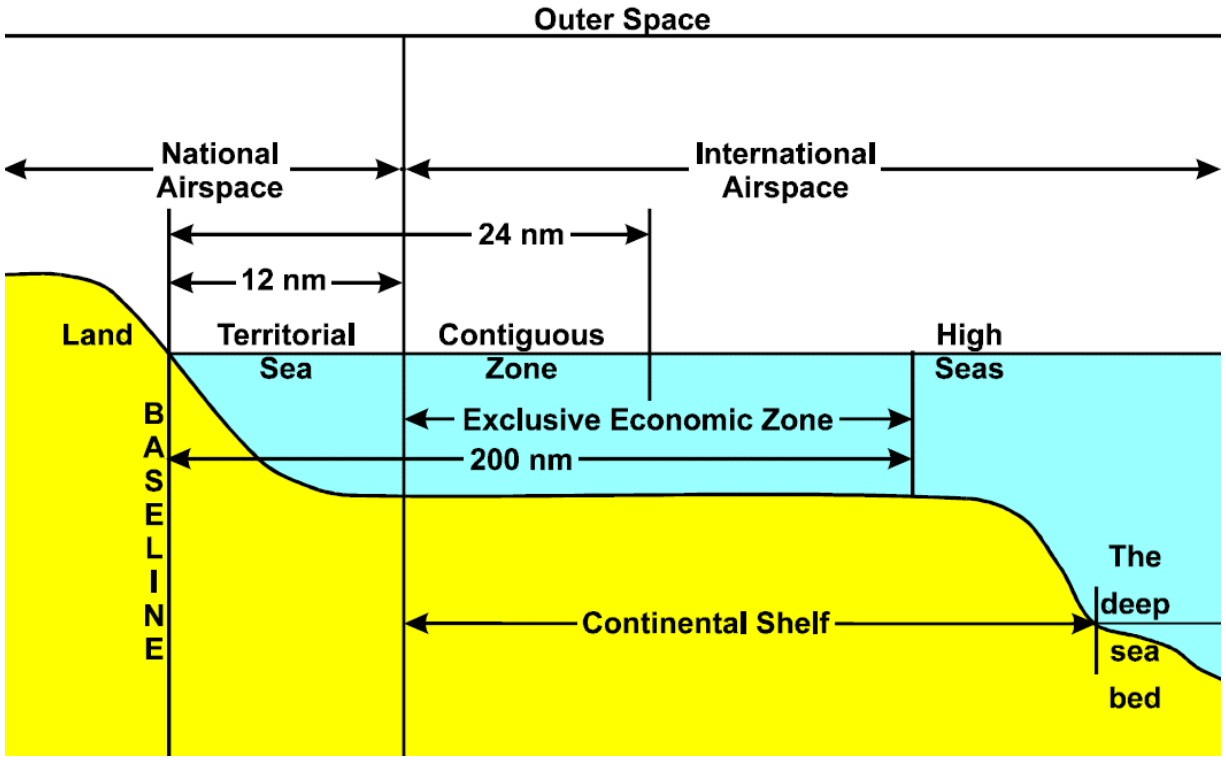
Fig. Maritime Area of Bangladesh

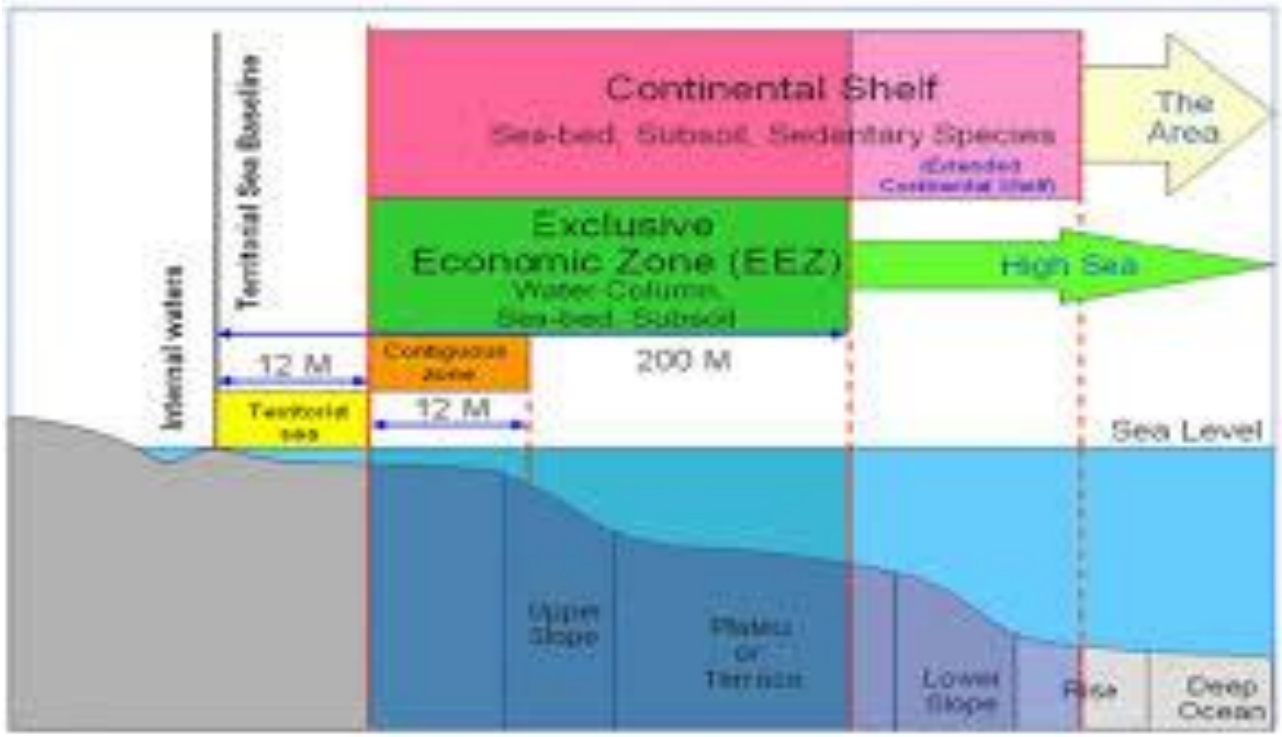


Bangladesh's Territorial Waters and Maritime Zone Act'1974

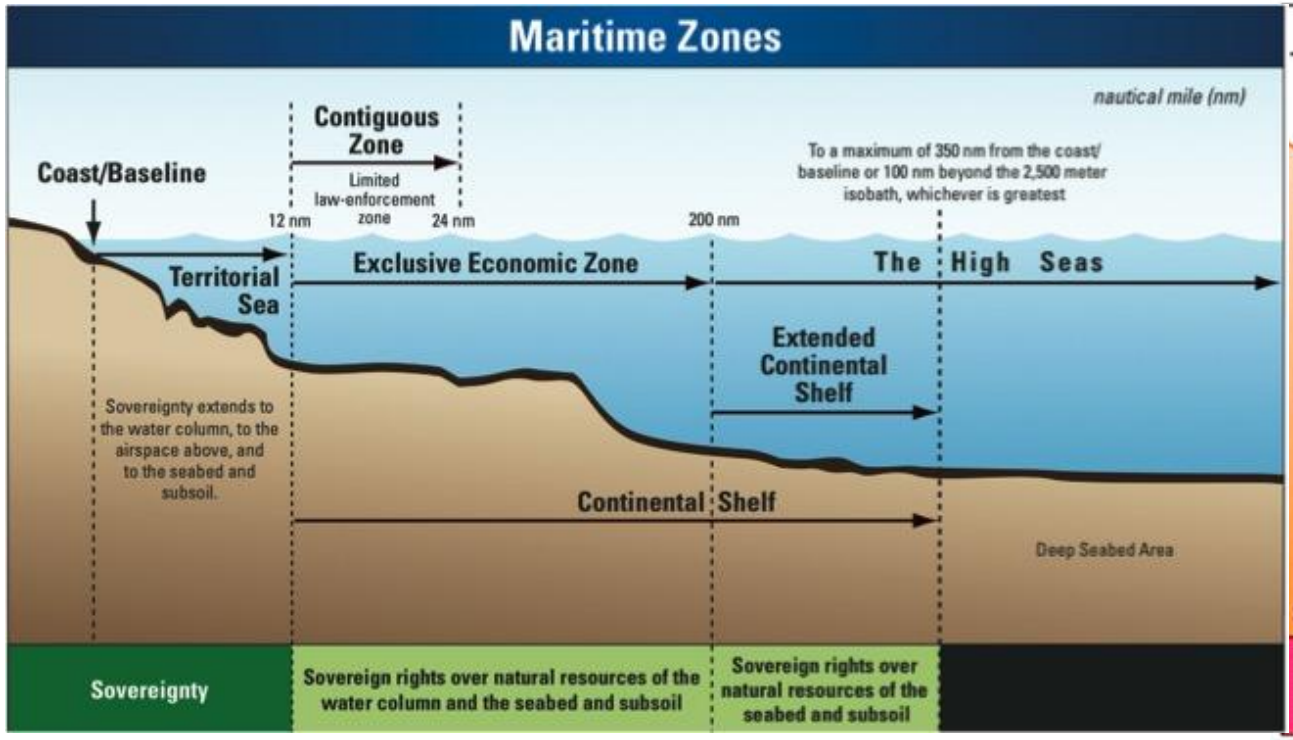


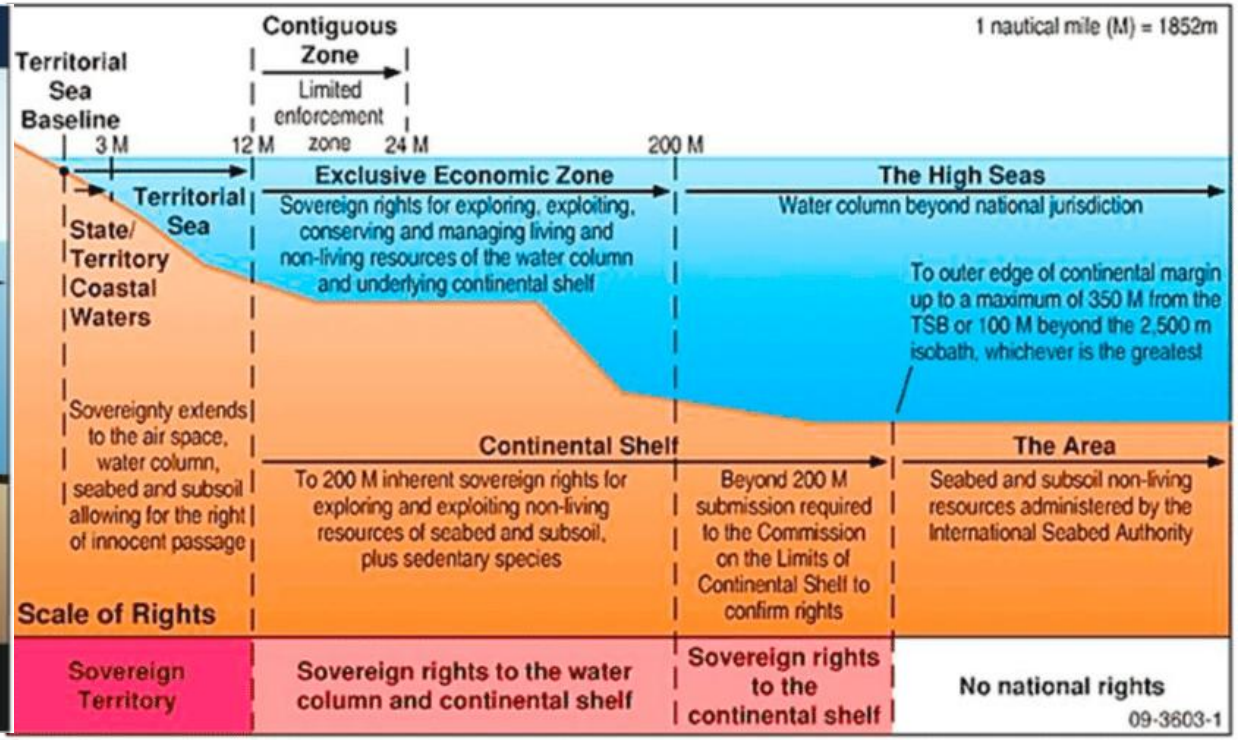
Maritime zones and rights under the UNCLOS 1982





Maritime zones and rights under the UNCLOS 1982





Conclusion

- ✓ There is no doubt that Bangladesh has gained **crucial economic benefit** in solving the **40 years** longstanding **maritime dispute.**
- ✓ The **verdict** will enable the enhancement of **medium and long- term energy security interests** of the country.
- ✓ With **expansion of the maritime boundary**, the opportunities for the **nation's fishing industry** have also been enhanced as fishing now can be carried out in the deep sea.
- ✓ As a **coastal state** now it can **explore and exploit** living and non-living resources of **water, seabed and subsoil of 200 nm exclusive economic zone**.
- ✓ At the same time it has **sovereign right** over its **350 nm continental shelf** in the sense that even if Bangladesh **does not exploit** any resources from that area, no one can do so **without previous permission** of the country.
- ✓ Bangladesh can get **relieve from its population pressure** utilizing its **marine resources properly**.
- ✓ For this, the country need to **take proper policy planning** otherwise it will **not be possible to reach its goal**.

Reference

- Churchill, R. (2012). The Bangladesh/Myanmar Case: Continuity and Novelty in the Law of Mantime Boundary Delimitation. Cambridge Journal of International and Comparative Law, 1, 137-152. http://dx.doi.org/10.7574/cjic 101.01.3
- Habibur Rahman, M. (1991). Delimitation of Maritime Boundaries (1st ed). Rajshahi: University of Rajshahi, 288.

❑ **Open the floor for questions and discussions**

THANK YOU